Scalable Real-Time Detection of AI-Generated Arabic Text

A Distributed Data Pipeline Approach using Apache Spark

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The Challenge: AI & Academic Integrity

The rapid proliferation of Large Language Models has introduced significant challenges to academic integrity, particularly concerning the undetected use of AI-generated text.

Scale

16,895 balanced Arabic academic abstracts

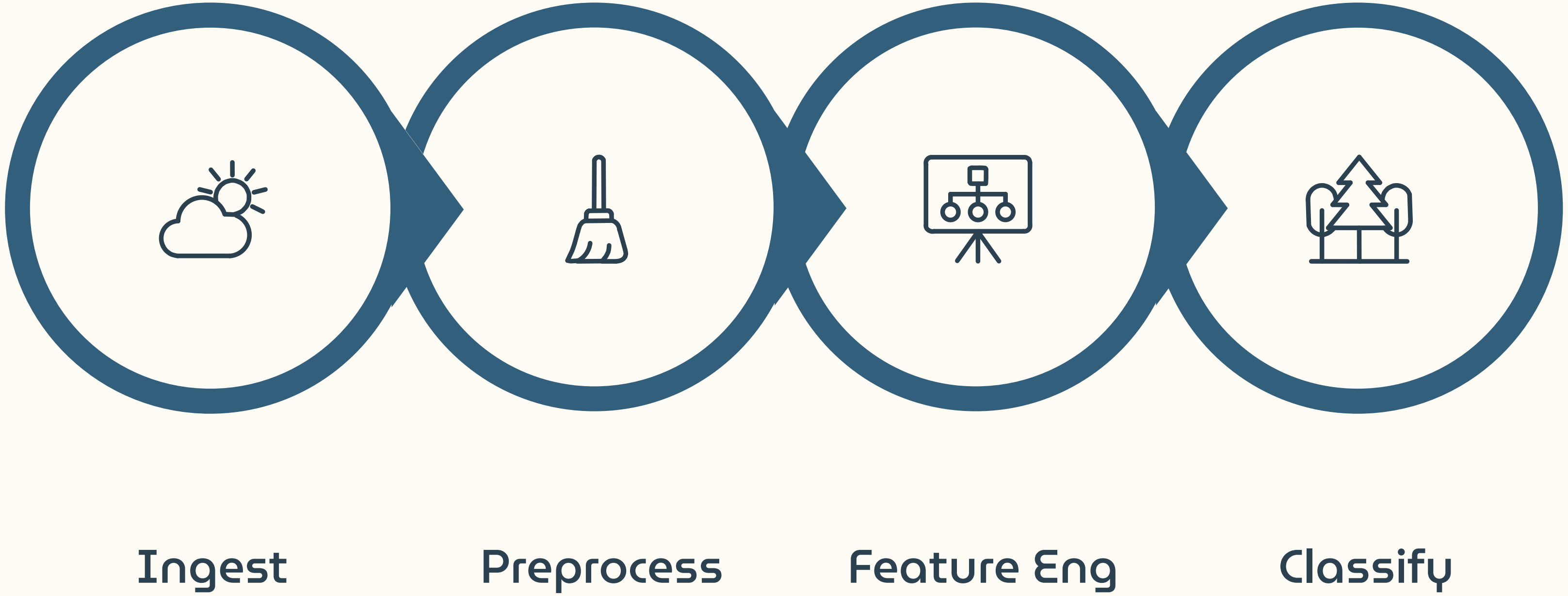
Goal

Distinguish human-written vs. AI-generated text

Platform

Apache Spark distributed pipeline

End-to-End Pipeline Architecture



The Apache Spark pipeline handles large-scale ingestion, processing, and real-time streaming — with Parquet persistence breaking the computational lineage for rapid iterative training.

Data Ingestion & Preprocessing

Dataset

16,895 balanced records from Hugging Face — evenly split between human-authored and AI-generated Arabic academic abstracts

Preprocessing

Diacritic removal, character normalization, and elimination of non-alphabetic symbols for a clean lexical baseline

Optimization

Intermediate DataFrames persisted to distributed **Parquet** format, breaking the Spark DAG and eliminating redundant processing

Dual-Track Feature Engineering

Lexical Features — TF-IDF

Top **2,000 contextual terms** extracted to capture the most discriminative vocabulary used by AI vs. human writers.

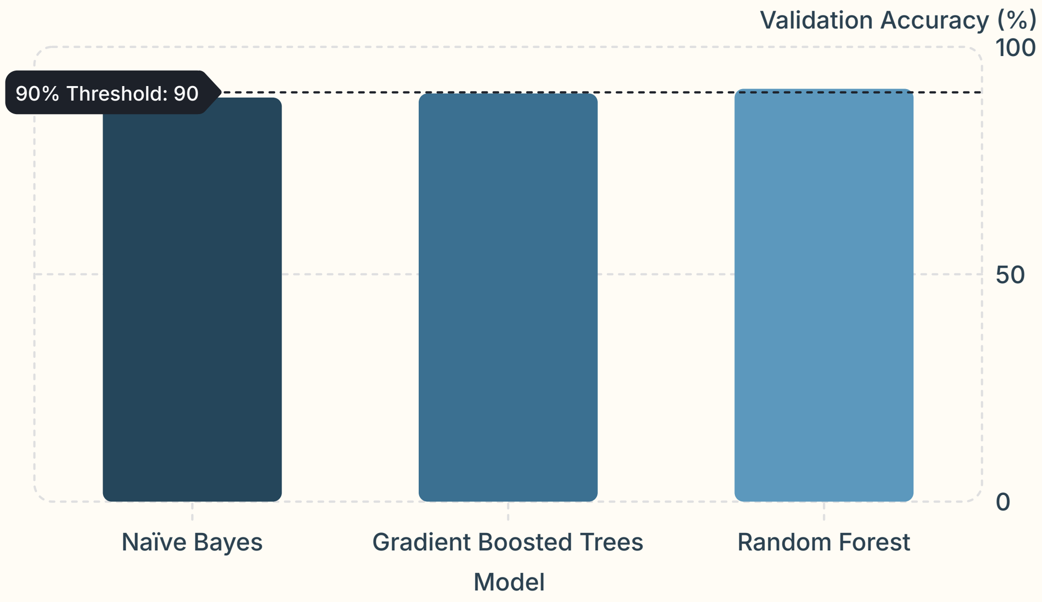
Stylometric Features — Custom UDFs

Five structural fingerprints engineered via **Camel Tools** morphological analyzer, executed as PySpark UDFs to capture syntactic writing style.

The Five Stylometric Features

1	**Word Length Variance (f15)** Human writers mix short and long words; AI uses uniform lengths.
2	**Chars per Paragraph (f38)** AI outputs equally sized paragraphs; humans vary by idea flow.
3	**Command Verb Count (f61)** Zero variance — academic abstracts are declarative, not imperative.
4	**Avg Sentence Length (f84)** **Most powerful feature** — AI sentences are mechanically consistent.
5	**Formality Score (f107)** Noun/adjective density vs. verbs/pronouns — AI tends overly formal.

Model Performance: Random Forest Wins



Champion Model: Random Forest

The only model to break the **90% accuracy threshold** at **90.78%**, demonstrating superior handling of the high-dimensional feature space without overfitting.

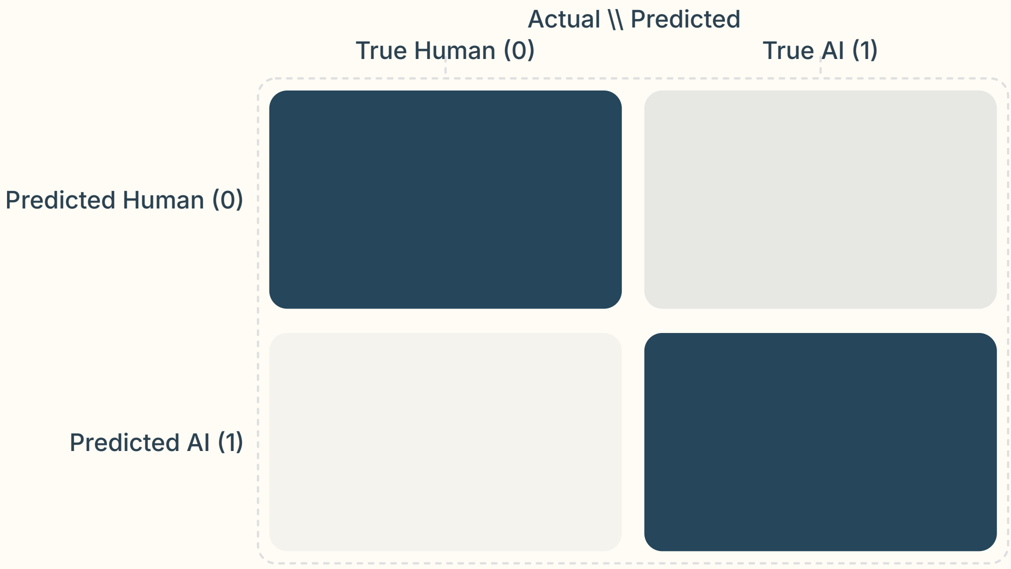
Train / Val / Test Split

70% / 15% / 15%

Baseline

Naïve Bayes at 88.86%

Confusion Matrix & Error Analysis

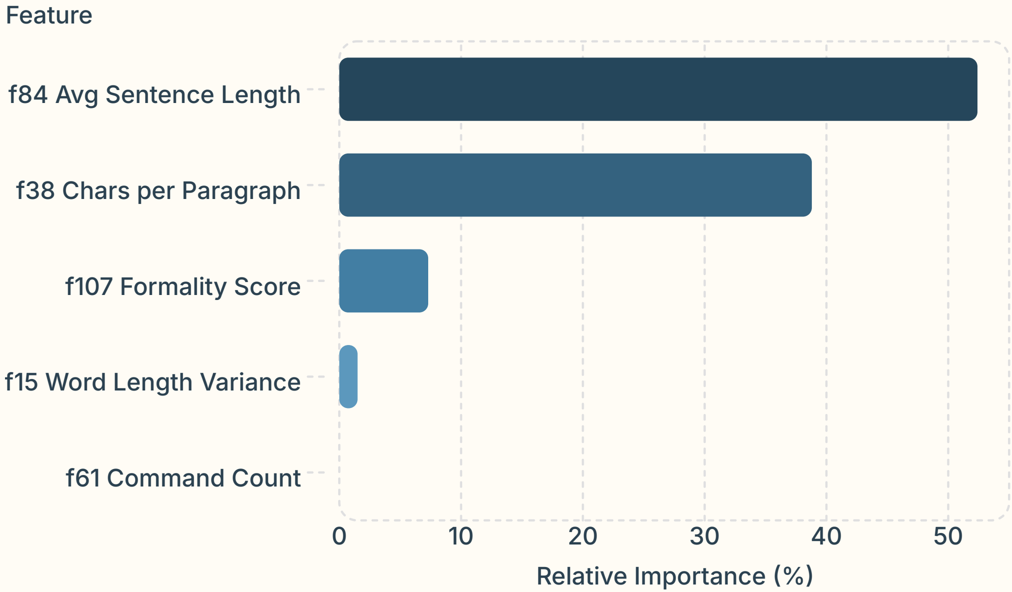


Balanced Classification Integrity

The model correctly identified **1,113 human** and **1,111 AI** abstracts with minimal class-propensity bias.

> Marginal errors were primarily **false positives** — human abstracts written in an excessively rigid, passive, and formal manner that mimicked LLM output patterns.

Feature Importance: The AI Footprint



Key Insight

AI models exhibit **highly standardized sentence structures**. **Average Sentence Length (f84)** alone accounts for **52.4%** of predictive power among custom features — the mechanical, consistent sentence lengths of AI contrast sharply with fluctuating human writing.

Command Count (f61) at **0.0%** confirms the system correctly recognized the declarative nature of academic text.

Real-Time Streaming & Conclusion

🔴 Live Pipeline

Spark Structured Streaming ingested and classified text files in real-time, processing mock messages at **5-second intervals** in a production-ready environment

🏆 Champion Result

Random Forest achieved **90.78% accuracy** on 16,895 balanced Arabic abstracts using TF-IDF + 5 custom stylometric features

🔭 Future Work

Expand to diverse disciplines, test against newer LLM iterations, and explore deep learning integration within the distributed Spark environment